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REVIEW ARTICLE

Impact of Active Antihyperglycemic Components as Herbal Therapy for Preventive Health Care Management of Diabetes

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Abstract: Diabetes is a metabolic hyperglycemic condition that progressively develops, effect small and large sensory fibers in the affected population. It has various complications as hypertension, coronary artery disease, stroke, blindness, kidney disease as well as peripheral neuropathy. Sulfonylureas, thiazolidinediones, metformin, biguanidine, acarbose and insulin are commonly used drugs for diabetic patients, but these all have certain side effects. Even metformin, that is known as the miracle drug for diabetes has been found to be associated with side effects, as during treatment it involves complications with eyes, kidneys, peripheral nerves, heart and vasculature. In the present article, we hypothesize recent discoveries with respect to active ingredients from Indian medicinal plants *i.e.,* polypeptide-p (protein analogue act as artificial insulin), charantin (a steroidal saponin), momordicin (an alkaloid) and osmotin (ubiquitous plant protein and animal analogue of human adeponectin) possessing anti-hyperglycemic potential for diabetes type II. Therefore, plants as herbal therapy have preventive care of hyperglycemia accompanied with healthy lifestyle which can provide significant decline in the incidences of diabetes in future.

Keywords: Diabetes, diabetic neuropathy, anti-hyperglycemic, preventive health care management, diacylglycerol, herbal therapy.

1. INTRODUCTION

Globally, 451 million people, aged 18 to 99 years were reported to be diabetic in the year 2017. It is also estimated that it can reach up to 693 million till 2045 [1]. Diabetes is a metabolic disease characterized by level of glycaemia. It occurs due to defect in secretion or action of insulin by the beta cells of pancreas. It may lead to microvascular or macrovascular complications resulting in problems of vision, kidney, heart, amputations and nerves [2]. Diabetes is classified into three types. Type I diabetes is insulin independent disease in which pancreas is destroyed due to autoimmune response of our body. Type II diabetes is insulin dependent, which occurs due to inefficient absorption of glucose into cell. Gestational diabetes develops during pregnancy due to some hormonal changes [3]. Diabetes depends on age, diet, genetic predisposition, obesity and lifestyle. It is a heterogeneous metabolic disorder recognized by hyperglycemia, glycosuria, and negative nitrogen balance resulting in various complications as diabetic-retinopathy, diabetic-nephropathy, diabetic-neuropathy, peripheral vascular disease, atherosclerosis, coronary artery disease, high cholesterol and high blood pressure *etc* [4]. Hyperglycemia activates diacylglycerol formation, due to which protein kinases C (PKC) are activated and in the presence of NADPH-oxidase it leads to the production of reactive oxygen species (ROS) elevating oxidative stress. Due to which angiogenesis is formed that modulates cellular death/apoptosis [5] (Fig. **1**). The scientific contributions sustaining the occurrence of insulin like hormones/proteins for the anti-hyperglycemic activity from plants. It has been estimated that one-third of the patients with diabetes use some form of complementary and alternative medicine. Plants gaining the most attention for their anti-diabetic potential as bitter melon, black plum, neem, tulsi and fenugreek possessing analogue of plant insulin or moeity enhancing the insulin sensitization [6]. Enormous pre-clinical research had been reported in antidiabetic/antihyperglycemic property of plants through different postulated mechanisms. However, clinical trial data for human subjects are inadequate due to faulty study design along with less statistical significance. The antidiabetic activity and pharmacological report on plant active metabolite to be used as a natural medicine as preventive health care against diabetes type II. This calls for optimized-designed clinical trials to further explicate its promising therapeutic impact on diabetes. Managing diabetes remains a challenge for both developing and developed countries. In some countries, implementation of evidence-based strategy and reformation of clinical

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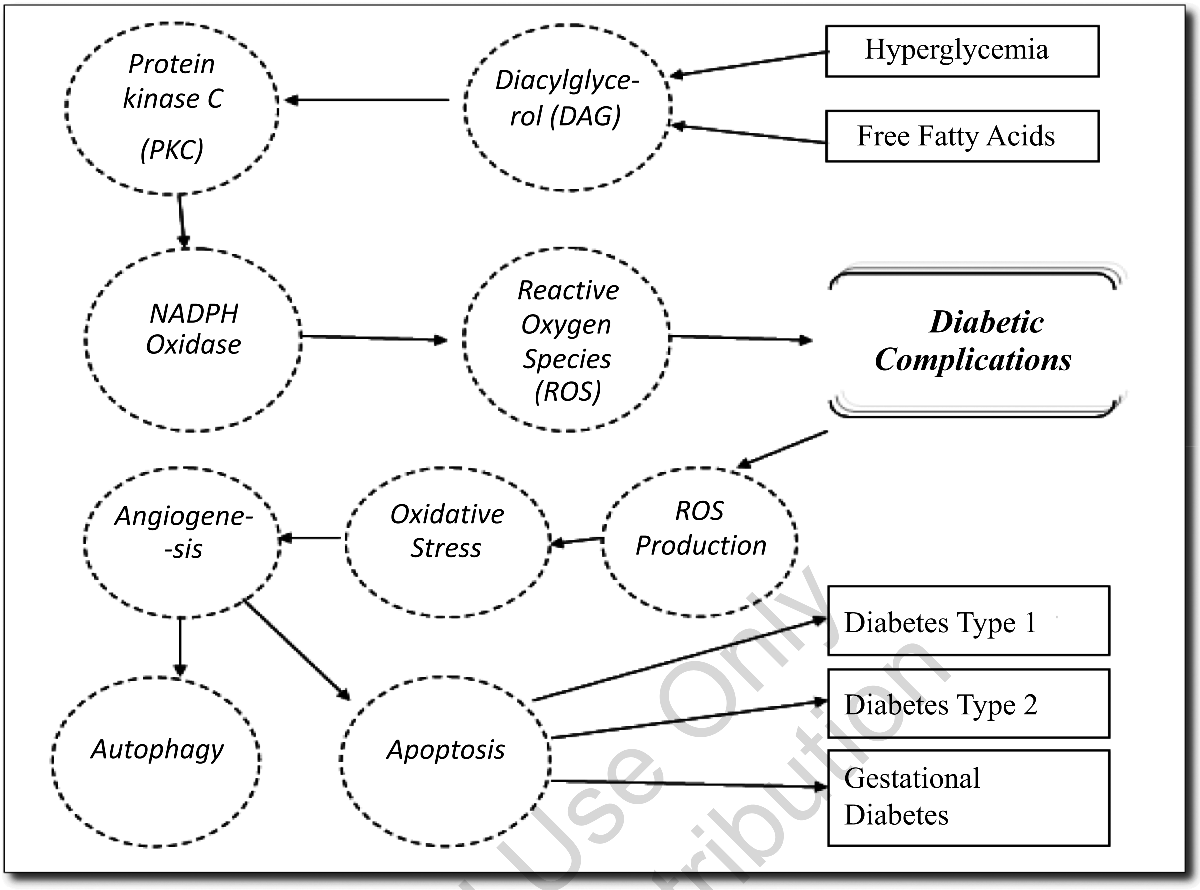

Fig. (1). A cascade of events in the development of diabetic neuropathy.

concern organization have proved to be significant. There have been numerous attempts in developing countries to generate realistic and effectual care systems [7]. Various plants used for the treatment of Diabetes were reported as *Eugenia jambolina, Azardica indica, Caesalpinia bonducella, Capparis decidua, Coccinia indica, Momordica charantia, Ocimum sanctum, Phylanthus amarus, Pterocarpus marsupium, Trigonella foenum graecum, Tinospora Cordifolia, Allium sativum* and *Withania somnifera* [8].

2. COMMONLY USED DRUGS FOR DIABETES

Various anti-hyperglycemic drugs are available in the market but their application is thoughtful due to associated adverse effects [9]. Therefore, there is an urgent need of alternative medication for diabetes since the existing treatment drugs have certain adversities as liver disease, fluid retention, weight gain, fracture risk, bladder cancer, ovulation, naso-pharyngitis, hives, urinary tract infections, facial swelling, stomach discomfort, diarrhea, decreased appetite, metallic taste, interference with vitamin B12 absorption, skin rash, drop in red blood cell count, liver disease, gas, bloating, vaginal or penis yeast infections, upper respiratory tract infections, kidney disfunction, hypersensitivity reactions, heartburn, nausea, haedache, weakness, respectively [10]. Recent evidence shows that metformin can cause lactic acidosis when it is taken regularly by the diabetic patients. Excess of sulfonulureas may lead to cardiovascular problems and hypoglycaemia [11]. Sulfonylureas, thiazolidinediones, and metformin are oral anti-diabetic drugs [12] which are currently in use. Thiazolidinediones and rosiglitazone have more risk of myocardial ischemic, congestive heart failure as well as fractures in women. Pioglitazone may cause fractures in women, weight gain, heart failure and bladder cancer. GLP-1 agonists and DPP-IV inhibitors may lead to pancreatitis or pancreatic cancer. Sodium-glucose cotransporter (SGLT) may cause urinary and vaginal infections. Insulin sometimes do excess of hypoglycaemia, cardiovascular or breast cancer [13]. Metformin, sulfonylureas, glinides, dpp-4 inhibitors, sglt-inhibitors, glp-1 receptor agonists, acarbose, pioglitazone, and insulin are the antihyperglycemic drugs used for the treatment of diabetes type II [14]. These drugs easily enter the hepatocytes by the action of organic cation transporter 1, whereas they are taken up in renal epithelium by the help of organic cation transporter 2 [15]. These reduce hyperglycemia by increasing peripheral glucose uptake and reducehepatic gluconeogenesis [16].

Treatment with these drugs along with healthy changes in life style can reduce the incidence of diabetes even in the people suffering from the high risk of diabetes, but none of these are safe and likely to produce side effects. Therefore, there is still a quest for the development of more effective anti-diabetic agents

[17]. Synthetic drugs analogous to sulphonylureas, biguanidine, acarbose and insulin are comprehensively used in Allopathic management of Diabetes. However, diabetes is known as silent killer and in recent times evidence of cases of "Insulin resistance" linked with the incidence of side effects from prolonged usage of conventional medicines have triggered the exploration for safe and effective alternatives.

3. DOMESTIC MEDICINES

A dietary approach to diabetes appears to be feasible to implement among diabetes type II [18]. Nutrition is a most important factor for preventive health care of diabetes. In our body, sugar damage and inflammations are cleared by serrapeptase, curcumin, vitamin D3 and ecklonia cava. In type II diabetes, alpha lipolic acid- R improves glucose metabolism. Antioxidants such as natural vitamin E (tocotrienol and tocopherolcomolex), krill oil, chromium, vitamins and minerals are also considered as preventives of diabetes. Dietary supplementation of *Trigonella foenum-graecum* modulates glucose homeostasis and prevents DM in prediabetics [19]. *Momordica charantia* has an anti-diabetic property. Indian people use various alternative medicines to treat diabetes traditionally [20]. It is found that a novel amino acid 4-hydroxyleucine is found in the seeds of *Trigonella foenumgraceum* (fenugreek), which stimulates insulin release by isolated islet cells in both rats and human. Hence its intake increases glucose metabolism. *Gymnema sylvester* is a climbing tree; it is believed that the extract of its leaves suppresses the elevation of blood glucose level by inhibiting glucose uptake by intestine [21]. *Abrus precautaris* along with the leaves of *Andrograp hispaniculata*, *Gymnema sylvestere* and seeds of *Syzygium cumini* also *Wattakaka volubilisstapf* are believed to be antidiabetic when its papery leaves are taken orally with cow's milk [22]. Extracts of *Aloe barbadensis* leaves also possess a hypoglycaemic effect [23]. It is believed to be glucose tolerant. Aqueous extract *Mangifera indica* (mango) shows a hypoglycaemic activity [24]. Root extract of *Tinospora cordifolia* (guduchi) lowers the glucose level in alloxan diabetic rats in six months [25]. *Acacia Arabica* (Babul) does not cure allaxanized animals but it has hypoglycaemic effect on the control animals group. Sucrose-fed rabbits are given aqueous homogenate of *Allium sativum* (garlic) orally. *Allium cepa* (onion) and *Ocium sanctum* (tulsi) are given to alloxan induced animals [26]. *Azadirachta indica* (neem) and *Momordica charantia* (bitter guard) are given to strepzotocin treated diabetic mice. *Momordica charantia* has prophylactic properties. Its fruit improves glucose tolerance as well as reduces the blood glucose level [27]. Therefore, an extensive research is much needed to explore antihyperglycemic component originated from plants.

4. NATUROPATHY FOR DIABETES

Around 60 % of world population believes in traditional medicines derived from medicinal plants. A naturopathic approach appears to be feasible to implement among diabetes type II [28]. Diabetes is recognized since ages and numerous treatments for the disease explored from the ancient times were based on the usage of plants and its parts. A study suggested that in old medical systems, like Ayurveda, make use of plants and herbs to combat diabetes. Diets of green vegetables, infusions, decoctions, steeped leaves or stems, or simply water extracts have been used since ages. The scientific origin for the consumption of most of these "natural medicines" has not been determined but dynamic principles have been isolated and identified exhibiting potent molecules of several chemical classes [29] detected the presence of salacinol, an α- glucosidase inhibitor extracted from *Salacia reticulate* roots, identification and successful usage of metformin, which was isolated from biguanides associated with the leaves of legume plant *Galega officinalis* and cryptolepine, anindoloquinolene alkaloid extracted from the leaves of *Cryptolepiss anguinolenta* [30]. "Natural medicines", based on plants continues to be used by poor populations globally. As indicated, the scientific mechanism for their utilization is known for a small number of them and the medical community more frequently do not believe these "natural medicines" for the management of the diabetes [31]. Through the present review, we wish to highlight the advantage of plant based therapy being acceptable by masses.

Preventive health care system and naturopathy play a considerable role in providing a platform to the researchers to pursue scientific interventions in Ayurveda, Naturopathy, offer a wide range of holistic treatments covering preventive, promotive, curative and rehabilitative and rejuvunatary needs [32]. These systems of medicine generally attract increasing attention globally. Naturopathy systems of medicine are being used for centuries and have continuous traditions of acceptance and practice. Treatment of Diabetes Type II can be possible by herbonatural component as possible drug candidates.

4.1. Plants as a Source of Insulin Production and Sensitization

The occurrence of insulin in plants is not acknowledged by the scientific community. This paradigm and reclaim information strongly suggests that insulin is certainly found to be associated with plants. On the basis of the review it was indicated that in leguminous plants, a protein molecule is present with the equivalent amino acid sequence as bovine insulin. Furthermore, evidences are observed that proteins associated with vertebrate's insulin signalling pathways are also detected along with insulin-like molecules in plants [33]. On the other hand, current and strong evidence proposes the presence of insulin in plants, suggesting that the hormone plays similar roles in plants as in metazoans [34].

4.2. Standardization

Plant-based medication has been used by approximately 30 % of the population and is found to be cost-effective for its treatment. In several parts of the world, particularly in low income countries, this may perhaps be the only form of therapy accessible to treat diabetic patients. There are several studies that report anti-diabetic potential of herbal plants [35]. Ayurveda and other conventional medicinal systems for the cure of diabetes also illustrate a number of herbal drugs [36]. Consequently, plants play an imperative role as alternative medicine due to fewer side effects and being economical. The active principles associated with the medicinal plants origin have been reported to possess pancreatic β cells regenerating, insulin sequestering and combating the crisis of insulin resistance. Insulin and oral hypoglycemic agents like sulphonylureas and biguanides are still the chief players in their cure and management, but there is a quest for the establishment of more efficient anti-diabetic agents. Several studies suggested that in certain conditions the utilization of some plant's active compound will be possible to add to curative and preventive health care management. Earlier information specifies that insulin ingested collectively with protease inhibitors is protected from hydrolysis in the elementary canal, crosses the intestinal barrier and supports the reduction of blood glucose levels [37]. Furthermore, the separation of a galactorhamnan polysaccharide in composite along with insulin in seed coats of Jack bean indicates that the hormone could be protected from hydrolysis in the digestive tract and induces lowering of blood sugar after passing the intestinal barrier. Ever since its discovery, comprehensive research, both academic and medically oriented, has recognized that insulin, besides playing its role as the conventional hormone coupled with the breakdown of glucose, also concerned with growth, reproduction and longevity processes [38]. Numerous components of the signaling pathway in which insulin is implicated have been recognized even though a complete understanding of their mechanism is unknown. Besides its traditional role in metabolic processes linking glucose and insulin has been discovered in both vertebrate and invertebrate [39]. There is an urge to find definitive procedure to be used as a preventive cure.

5. ACTIVE INGREDIENTS FROM PLANTS AGAINST DIABETES

Although various drugs for diabetic patients are reported, none of them have the potential to recover diabetes until now [40]. Metformin, sulfonylureas, glinides, DPP-4 inhibitors, SGLT-inhibitors, GLP-1 receptor agonists, acarbose, pioglitazone, and insulin are the antihyperglycemic drugs used for the treatment of type II diabetes [41]. Treatment with metformin and changes in life style can reduce the incidence of diabetes even in people suffering from the high risk of diabetes but none of these are safe and likely to produce side effects. Therefore, there is a need for the development of more effective anti-diabetic agents. Metformin an established drug for diabetes also has side effects [10]. In spite of the known fact that ayurveda and naturopathy are successful for hyperglycemia but in the current scenario, we have limited success stories with their use in human. Scientists are extensively working on the use of plants to fight diabetes but clinical trial data for human subjects are limited due to some flaws in research design along with little statistical relevance [42]. The antihyperglycemic activity and pharmacological description on plant active metabolite must be used as a natural medicine as preventive health care for diabetes type II and calls for better-intended clinical trials to further elucidate its possible therapeutic effects on diabetes [43].

Antihyperglycemic components isolated from test plants will be more efficient than allopathic cure as they have no side effects, it can be recommended to be used alone or in combination with fewer medicines. Such reports give us impetus to explore and analyze action of these components under *in vivo* study to observe the bio efficacy of these components in animal model system. Scientific contributions support the presence of insulin like hormones/proteins for bioactivity from plants. That can be potentially being utilized by society at large as a preventive and curative approach of medication for Diabetes II.

5.1. Plant Analogue of Insulin

Extract of the unripe fruits of *Momordica charantia* is significant in the management of type II diabetes. It consists of insulin-like hypoglycemic protein polypeptide-p which lowers blood glucose levels in gerbils, langurs, and humans when injected subcutaneously [44]. It works by mimicking the action of insulin in the body. Bitter melon contains polypeptide-p used to control diabetes naturally by acting as an artificial plant insulin. Polypeptide-p or p-insulin is an insulin-like hypoglycemic protein, revealed to reduce blood glucose levels in humans when injected subcutaneously. The p-insulin mechanism is through mimicking the action of human insulin acting as its analogue in the body and consequently be used as plant-based insulin substitution in patients with type I diabetes. Polypeptide-p is a reported antidiabetic compound which is extracted from bitter melons [45]. Polypeptide-p can be isolated from fruits and seeds of plant *Momordica charantia*. It has the antihyperglycemic activity which lowers the blood glucose level. Polypeptide-p promotes balanced storage of glucose in the liver as glycogen, supports healthy insulin secretion by the islets of Langerhans, maintains peripheral glucose utilization, and healthy serum protein levels according to *in vivo* and *in vitro* studies [46]. In the presence of protein kinase B (PKB, or Akt) polypeptide-p can regulate intracellular insulin signaling, which helps in cell metabolism, growth, proliferation, and survival [47]. Glycogen is synthetized and glucose transportation is regulated (Fig. **2**).

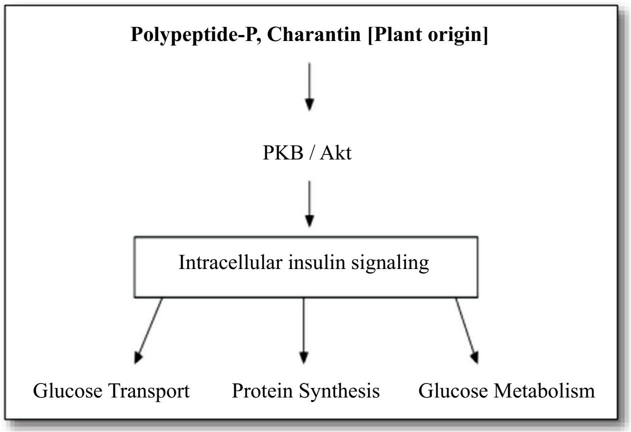

Fig. (2). Role of polypeptide-P as artificial insulin or insulin analogue.

5.2. Animal Analog of Adiponectin

A plant-derived component osmotin is a ubiquitous plant protein that plays an important role to mimic the hormone that mitigates diabetes and neuro-degenerative disorders. Osmotin occurs in two forms: osmotin-I (aqueous soluble form) and osmotin-II (detergent soluble) in the ratio of 2:1. Osmotin I and osmotin II, both exhibit antidiabetic activity. Cultured cells of *Nicotiana tabacum* are reported to synthesize osmotin [48]. Osmotin protects rat foetal hippocampal neurons as well as neonatal hippocampal and cortical neurons *in vivo* against ethanol-induced apoptosis. Osmotin attenuates amyloid beta-induced memory impairment, tau phosphorylation and neurodegeneration in the mouse hippocampus [49]. Since, osmotin has important roles in protection against stress conferring conditions in the brain and it is beneficial in the brain too. Therefore, osmotin and polypeptide-p can also be effective in natural preventive therapy against the diabetic neuropathy. Adiponectin is an adipokine that is specifically and abundantly expressed in adipose tissue and directly sensitizes the body to insulin [50]. Up- regulation of adiponectin is a partial cause of the insulin- sensitizing and antidiabetic action of thiazolidinediones. Therefore, adiponectin and its receptors represent potential versatile therapeutic targets to combat obesity-linked diseases characterized by insulin resistance. Conversely, osmotin acts similar to the mammalian hormone adiponectin in a variety of *in vitro* and *in vivo* models [51]. Adiponectin and osmotin, the two receptor binding proteins do not possess sequence analogy at the amino acid level, but surprisingly they have similar structural and functional properties [52]. In adipose tissue glucose uptake, triglyceride synthesis and insulin secretion are regulated when 5' Adenosine Monophosphate-activated Protein Kinase (AMPK) and Sirtuin 1 (SIRT1) pathways are regulated. Leading to the regulation of Peroxisome Proliferator-activated Receptor Gamma Coactivator 1-alpha (PGC-1α), a molecule expressed in brown adipocytes [53] (Fig. **3**).

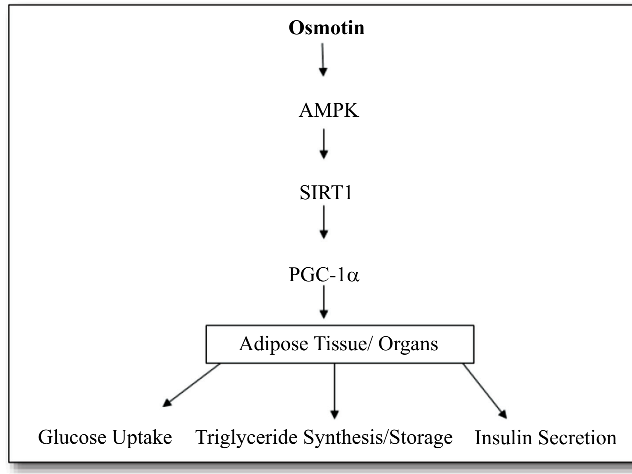

Fig. (3). Osmotin: as animal analogue of adeponectin.

5.3. Charantin from Plant Source

Charantin is a cucurbitane-type triterpenoid in bitter gourd and fenugreek with antidiabetic potential. This indicates that charantin is responsible for the hypoglycemic activity that can be potentially utilized against diabetics. Unripe fruit, seeds and aerial parts of *Momordica charantia* and *Trigonella foenum-graecum* are used from long time worldwide to treat diabetes. It reduces the fasting blood glucose and also improves glucose tolerance in normal and diabetic animals as well as humans. It can be taken orally in the form of juice or powder [54].

6. *IN VIVO* STUDY OF ACTIVE ANTIHYPER-GLYCEMIC BIOMOLECULES

The formulation of the most effective herbalo natural cure can further be tested for *in vivo* study in different model system for its viability and efficacy. The conjugate can be suggested for multiple level trials to result in translational product [55]. These components can be further tested in the cell lines/ animal model system to check the bioavailability and mechanism of action of the antihyperglycemic component. The pPrepared formulation may be a consortium of active moieties as a natupropathy for hyperglycemia.

The concept of food as medicine is a focal theme in nutritional and dietetic sciences. Plants have been used as a preventive health care system in the form of dietary supplements and ethno medicine from centuries for the cure of diabetes. Bitter gourd, fenugreek, wheat, potato has been comprehensively studied globally for their medicinal behavior. They are demonstrated as a versatile plant worthy of treating many diseases [56].

Although plentiful data from biochemical and animal studies areis available but the clinical data as reviewed was found to be often flawed by small sample size, lack of control and poor study designs. There is a need for the better-designed pre-clinical trials with sufficient sample size and statistical power to further indicate the acclaimed efficacy of plant's active components as a natural nutritional treatment for diabetes. In particular, may be a feasible option for ethnic minorities who have a high prevalence of diabetes but prefer treatment based on natural products according to their cultural beliefs.

6.1. Animal Study

Induction and establishment of diabetes in the experimental animal during *in vivo* studies are incorporated by using alloxan and streptozotocin all over the world. Superoxide radicals are formed by the alloxan and its reduction product dialuric acid due to the formation of a redox cycle. By the Fenton reaction highly reactive hydroxyl radicals are formed, after these radicals undergo dismutation to hydrogen peroxide [57]. Rapid destruction of β cells is caused by the action of reactive oxygen species with a simultaneous massive increase in cytosolic calcium concentration. Streptozotocin causes alkylation of DNA by entering in the B cell *via* a glucose transporter (GLUT2) [58]. Activation of poly ADP-ribosylation is induced by the damage of DNA, a process which is more important for the diabetogenicity of streptozotocin than DNA damage itself. As a result, hydrogen peroxide and hydroxyl radicals are generated. Due to the action of strepzotocin, β cells undergo necrosis and toxic amounts of nitric oxide are liberated that inhibits aconitase activity and participates in DNA damage [59], using various doses and routes of administration. To the experimental animals, diabetes can also be caused by the removal of pancreas, diabetogens, virus or transcription factor. Specifically, genetic engineered rodent models have revolutionized research in diabetes as experiments in human are limited and also naturally occurring [60]. Alloxan and streptozotocin get accumulated in the pancreatic β cells via the GLUT2 glucose transporter as they are toxic glucose analogues. In a cyclic redox reaction with its reduction product in the presence of intracellular thiols such as glutathione alloxan generates reactive oxygen species (ROS) [61]. Superoxide radicals, hydrogen peroxide and in a final iron-catalysed reaction step hydroxyl radicals are produced as a result of autoxidation of dialuric acid. These hydroxyl radicals have a particularly low antioxidative defence capacity consequently death of the β cells occur [62].

Juvenile onset diabetes was reported in 1960s by Gamble and co-workers. Reports suggested that that diabetes can be induced by viruses named D-variant of encephalomyocarditis (EMC-D), which develops insulin dependent hyperglycemia by destroying beta cells of pancreas whereas Coxsackie virus also develops insulin dependent diabetes by destroying acinar cells of pancreas [63]. Repeated administration of growth hormone and Corticosteroid can lead to stable or irreversible diabetes as it develops ketonuria and ketonemia [64, 65]. A diet with high fat content with water ad libitum is also used to make mice diabetic as it causes obesity, hyperinsulinaemia and altered glucose homeostasis [66]. Pancreatectomy is a surgical method to induce diabetes in laboratory mice. It causes hyperglycaemia and insulin resistance but not obesity [67]. It requires well sanitized surgical room as well as technical expertise.

Working on the animal model, helps us to know about a disease, its pathophysiology and also in the development of drugs for its treatment. According to the data of World Health Organization (WHO), there are approximately 160,000 diabetics globally, the number of diabetic patient has doubled in the last few years and is expected to double once again in the year 2025, so it is a medical concern [68]. Animal studies have constantly presented the hypoglycemic impact of bitter melon, fenugreek, wheat, potato and its plant part as leaves, seeds, fruit and pulp in various reports [69, 70]. The studies have revealed that an assortment of forms of these plants has the prospective to improve glucose tolerance, subdue the postprandial activity in hyperglycemia in experimental mice and augment insulin sensitization and lipolysis. Various human studies have expressed comparable hypoglycemic effects of these plants [71]. These plants can suggest a better approach toward safe treatment subdue the adversities associated with existing treatment methodologies for a prolong treatment.

6.2. Dosage and Safety

As a dietary supplement, it is frequently sold as an extract, powder and juiced. All parts of the plant are used alone or in consortia. The inadequate standardization may also increase the incidence of undesirable side effects. Hypoglycemia and hepatotoxicity have been demonstrated by a few studies. Bitter melon administration may produce an additional effect ion humans when used along with insulin or any other hypoglycemic drug [72].

6.3. Novelty Content

Based on the review searched extensive work had been reported on the significance of the plants in the treatment of both diabetes I and II, no definite formulation is commercially available except metformin (originated from plant metabolite). Prolonged therapy with metformin has some side effects [73, 74]. Therefore, to present some biomolecules as artificial insulin or its analogue and osmotin protein (analogue of adeponectin) that increases the sensitization of existing amount of insulin and proves to be a boon for diabetes type II patients. Previous reports provide inadequate information regarding the *in vivo* trials and devoid of statistical efficacy determination. Therefore, scientists must try for the potential biosimilars possessing insulin like properties for type I or enhancing the insulin sensitization for type II diabetic patients.

CONCLUSION

Currently, type II Diabetes (Insulin-independent) is treated by prolonged allopathic medication which leads to the onset of type I Diabetes (Insulin dependent) for long term usage along with severe impacts on physiology of various organs. In this context, scientists and researchers need to search for an effective medicine or preventive health care management system for the welfare of the affected population, globally. There is an urgent need to explore a tool that increases the insulin sensitization in case of incidence of diabetes II as well as bioactive molecule possessing insulin like property. According to the present review comprehending the implications associated in the current context, for the treatment of diabetes scientists are now relying on active ingredients isolated from the plant parts. Osmotin is one such compound which acts as a plant analogue of adeponectin and thaumatin found to be associated with tobacco, wheat, potato and strawberry. It mimics the role of hormone which mitigates diabetes and neurodegenerative disorder. Polypeptide-p, charantin act as artificial insulin or analogue for insulin, promotes balanced storage of glucose or regulate intracellular insulin signaling associated with bitter melon and fenugreek. Here we propose the utilization of osmotin, Polypeptide-p and charantin to be explored as a potential treatment strategy against diabetes type II, hitherto unreported.

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

Disclosure of potential conflicts of interest indicates that there is no conflict of interest during the study conductin any of the authors. The study is conducted in compliance with the Ethical Standards.

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